Status	Finished
Started	Wednesday, 1 April 2026, 11:45 AM
Completed	Wednesday, 1 April 2026, 12:12 PM
Duration	26 mins 53 secs

Question 1

Complete

Marked out of 1.00

If the gradient at a point is zero, it indicates:

Select one:

- a. Always maximum
- b. Always minimum
- c. Could be max, min, or saddle point
- d. No solution

Question 2

Complete

Marked out of 1.00

In a chess AI, the system evaluates possible opponent moves before deciding its own. Which concept is applied?

Select one:

- a. Constraint satisfaction
- b. Adversarial search
- c. Clustering
- d. Linear regression

Question 3

Complete

Marked out of 1.00

During PCA, eigenvectors with the highest eigenvalues are selected. Why?

Select one:

- a. They reduce computation
- b. They represent maximum variance
- c. They normalize data
- d. They remove noise

Question 4

Complete

Marked out of 1.00

Gradient represents:

Select one:

- a. Function value
- b. Rate and direction of maximum increase
- c. Minimum value
- d. Constant

Question 5

Complete

Marked out of 1.00

A company processes real-time streaming data from millions of users. Which Big Data characteristic is MOST relevant?

Select one:

- a. Variety
- b. Velocity
- c. Volume
- d. Value

Question 6

Complete

Marked out of 1.00

A self-driving car detects obstacles using sensors and adjusts speed accordingly. Which type of agent is this?

Select one:

- a. Simple reflex agent
- b. Model-based agent
- c. Goal-based agent
- d. Utility-based agent

Question 7

Complete

Marked out of 1.00

A company wants to group customers based on purchasing behavior without predefined labels. Which approach is appropriate?

Select one:

- a. Supervised learning
- b. Reinforcement learning
- c. Unsupervised learning
- d. Semi-supervised learning

Question 8

Complete

Marked out of 1.00

A chatbot successfully convinces 70% of evaluators that it is human in text conversation but fails in reasoning tasks. What does this imply?

Select one:

- a. It passes strong AI criteria
- b. It passes Turing Test but lacks general intelligence
- c. It is an expert system
- d. It is unsupervised learning

Question 9

Complete

Marked out of 1.00

"All employees must wear ID cards." "Pratik is an employee." What inference technique derives "Pratik must wear an ID card"?

Select one:

- a. Induction
- b. Abduction
- c. Deduction
- d. Heuristic

Question 10

Complete

Marked out of 1.00

A healthcare AI system consistently underdiagnoses a disease in a specific demographic group due to skewed training data. Which principle is MOST directly violated?

Select one:

- a. Transparency
- b. Fairness
- c. Accountability
- d. Robustness

Question 11

Complete

Marked out of 1.00

Bias in AI systems primarily arises from:

Select one:

- a. Hardware limitations
- b. Training data
- c. Algorithms only
- d. User interfaces

Question 12

Complete

Marked out of 1.00

Which principle ensures AI decisions are understandable?

Select one:

- a. Fairness
- b. Transparency
- c. Accountability
- d. Privacy

Question 13

Complete

Marked out of 1.00

In a recommendation system, cosine similarity is used. What does it measure?

Select one:

- a. Distance between vectors
- b. Angle between vectors
- c. Magnitude difference
- d. Vector sum

Question 14

Complete

Marked out of 1.00

A loss function has multiple local minima. Which technique helps avoid getting stuck?

Select one:

- a. Batch gradient descent
- b. Stochastic gradient descent
- c. Deterministic search
- d. Linear regression

Question 15

Complete

Marked out of 1.00

Expert systems are based on:

Select one:

- a. Neural networks
- b. Rule-based reasoning
- c. Genetic algorithms
- d. Reinforcement learning

Question 16

Complete

Marked out of 1.00

A robot must find the shortest path in a maze with equal step costs. Which algorithm guarantees optimality?

Select one:

- a. Depth-First Search
- b. Greedy Search
- c. Breadth-First Search
- d. Hill Climbing

Question 17

Complete

Marked out of 1.00

State-space representation is used for:

Select one:

- a. Data storage
- b. Problem solving
- c. Visualization
- d. Optimization

Question 18

Complete

Marked out of 1.00

Scheduling exams without time conflicts is an example of:

Select one:

- a. Classification
- b. CSP (Constraint Satisfaction Problem)
- c. Clustering
- d. Regression

Question 19

Complete

Marked out of 1.00

A facial recognition system fails in low lighting conditions. Which limitation does this highlight?

Select one:

- a. Ethical bias
- b. Data dependency
- c. Lack of supervision
- d. Poor optimization

Question 20

Complete

Marked out of 1.00

A medical diagnosis system uses IF-THEN rules to infer diseases. What type of system is this?

Select one:

- a. Neural network
- b. Expert system
- c. Genetic algorithm
- d. Reinforcement model

Previous activity

Mid Quiz

Jump to...

Next activity

End Module Exam